

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

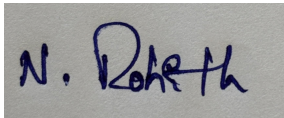
Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I **NALIPI REDDY ROHITH/192425115**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Error Analysis Task

Question

A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Solution

Introduction

A **Neural Network** is a machine learning model that learns patterns from training data and predicts outcomes for new data. The performance of a regression model is commonly measured using **Mean Squared Error (MSE)**.

The formula for MSE is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of observations

A **lower MSE** indicates better prediction accuracy.

Given:

- Training MSE = **0.85**
- Testing MSE = **5.20**

The testing error is much higher than the training error, which indicates that the model is not performing well on unseen data.

Analysis of the Results

Training Phase

The training MSE is **0.85**, which is very low. This means the model has learned the training dataset effectively and makes accurate predictions on the training data.

Testing Phase

The testing MSE is **5.20**, which is significantly higher. This means the model performs poorly when new student data is given.

The large difference between training and testing errors indicates **overfitting**.

Possible Reasons for the Difference

1. Overfitting

The model has memorized the training data instead of learning the general relationship between input features and student scores.

As a result:

- Training error becomes very low.
- Testing error becomes very high.

This is the most common reason for the given results.

2. Small Training Dataset

If only a limited number of student records are used for training, the model cannot learn all possible patterns.

A larger dataset generally improves the model's ability to generalize.

3. Complex Neural Network

If the network contains too many hidden layers or too many neurons, it becomes unnecessarily complex.

A highly complex model can memorize the training data, leading to poor performance on testing data.

4. Poor Data Quality

Training data may contain:

- Missing values
- Incorrect student marks
- Duplicate records
- Outliers

These issues affect the learning process and reduce prediction accuracy.

5. Irrelevant Features

Features such as:

- Student ID
- Roll Number
- Registration Number

do not contribute to predicting student scores. Including such features increases model complexity without improving accuracy.

6. Lack of Regularization

If techniques such as **L1**, **L2 Regularization**, or **Dropout** are not used, the model can easily overfit the training data.

7. Excessive Training

Training the model for too many epochs allows it to memorize the training samples.

Initially, both training and testing errors decrease, but after a certain point, the testing error starts increasing while the training error continues to decrease.

Corrective Measures

1. Increase Training Data

Using more student records helps the model learn better patterns and reduces overfitting.

2. Apply Regularization

Use:

- L1 Regularization
- L2 Regularization

These techniques reduce the complexity of the model by limiting large weight values.

3. Use Dropout

Dropout randomly disables some neurons during training.

Benefits:

- Prevents memorization.
- Improves generalization.
- Reduces overfitting.

4. Early Stopping

Monitor the validation error during training.

Stop training when the validation error begins to increase.

This prevents unnecessary training.

5. Normalize the Data

Scaling the input features using **Standardization** or **Min-Max Normalization** improves convergence and stability during training.

6. Reduce Network Complexity

Decrease:

- Number of hidden layers
- Number of hidden neurons

A simpler model often performs better on unseen data.

7. Feature Selection

Remove unnecessary features and keep only useful ones such as:

- Attendance
- Assignment marks
- Internal marks
- Study hours

This improves prediction accuracy.

8. Cross-Validation

Use **k-fold cross-validation** to evaluate the model on different subsets of the data.

This provides a more reliable estimate of model performance.

Comparison of Results

Dataset	MSE	Interpretation
Training	0.85	Good learning on training data
Testing	5.20	Poor performance on unseen data
Difference	4.35	Indicates overfitting

Expected Improvement

After applying the corrective measures:

- Training MSE may increase slightly.
- Testing MSE should decrease significantly.
- The gap between training and testing errors becomes much smaller.
- The model generalizes better to new student data.

Conclusion

The neural network achieves a **training MSE of 0.85** but a **testing MSE of 5.20**, indicating that it has learned the training data well but fails to generalize to unseen data. The primary reason is **overfitting**, which may be caused by a complex model, insufficient training data, poor feature selection, noisy data, or lack of regularization. By applying techniques such as **regularization, dropout, early stopping, feature scaling, feature selection, and cross-validation**, the model's testing performance can be significantly improved, resulting in better prediction accuracy and reduced error on new data.